

CONTACT	Email: kameron.harris@wwu.edu Web: glomerul.us Code: kamdh , glomerulus-lab
POSITIONS	Associate Professor, 2025–present Department of Computer Science, Western Washington University, Bellingham, Washington. Assistant Professor, 2020–2025 Department of Computer Science, Western Washington University, Bellingham, Washington. Washington Research Foundation Postdoctoral Fellow, 2018–2020 Paul G. Allen School of Computer Science & Engineering and Department of Biology University of Washington, Seattle, Washington.
RESEARCH INTERESTS	I am an applied mathematician and computer scientist with an interest in computational neuroscience , machine learning , and networks/graph theory . I enjoy collaborative, interdisciplinary research at the intersection of theory, modeling, and data science.
PUBLICATIONS	A list of publications including citation estimates and metrics is on Google Scholar. Key: * = WWU undergrad, † = WWU grad student, ‡ = other student mentee. 32. R Preble[†], P Venkatesh, S Mihalas, KD Harris. Neural Variability Enhances Artificial Network Robustness. Under review. 31. Veronica Hahn*, A Read[†], S Ban*, R Gupta*, KD Harris. Visual cortex-inspired compressive sensing. In preparation for submission. 30. R Osuna-Orozco, E Castillo, KD Harris, SR Santacruz. Identification of Recurrent Dynamics in Distributed Neural Populations. PLoS Comp Bio 21(2): e1012816, 2025. 29. V White*, M Chaudhary[‡], G Wolf, G Lajoie, KD Harris. Learning Stochastic Rainbow Networks. NeurIPS SciForDL workshop, 2024. 28. V White*, M Chaudhary[‡], G Wolf, G Lajoie, KD Harris. Learning and Aligning Structured Random Feature Networks. ICLR Re-Align workshop, 2024. 27. GM Chou*, NE Bush, RS Phillips, NA Baertsch, KD Harris. Modeling effects of variable preBötzing complex network topology and cellular properties on opioid-induced respiratory depression and recovery. eNeuro 22 January 2024, ENEURO.0284-23.2023. 26. M Xie[‡], S Muscinelli, KD Harris, A Litwin-Kumar. Task-dependent optimal representations for cerebellar learning. eLife 12:e82914, 2023. 25. KD Harris, O López, A Read*, Y Zhu. Spectral gap-based deterministic tensor completion. In proceedings of SampTA, 2023. 24. S Daetwiler*, A Read*, J Stillwell*, KD Harris. BrainViewer: interacting with spatial connectome data at the mesoscale. Preprint, 2022. 23. B Pandey[‡], M Pachitariu, BW Brunton, KD Harris. Structured random receptive fields enable informative sensory encodings. PLoS Comp Bio 18(10): e1010484, 2022. 22. KD Harris, Y Zhu. Deterministic tensor completion with hypergraph expanders. SIAM Mathematics of Data Science 3(4), 1117–1140, 2021. 21. KD Harris, A Aravkin, R Rao, BW Brunton. Time-varying Autoregression with Low Rank Tensors. SIAM Applied Dynamical Systems 20(4), 2335–2358, 2021. 20. G Brito, I Dumitriu, KD Harris. Spectral gap in random bipartite biregular graphs and applications. Combinatorics, Probability, & Computing, 1–39, 2021. 19. WS DeWitt, KD Harris, AP Ragsdale, K Harris. Nonparametric coalescent inference of mutation spectrum history and demography. Proceedings of the National Academy of Sciences 118 (21) e2013798118, 2021. 18. SM Hirsh[‡], KD Harris, JN Kutz, BW Brunton. Centering Data Improves the Dynamic Mode Decomposition. SIAM Applied Dynamical Systems 19(3), 1920–1955, 2020. 17. KD Harris. Additive function approximation in the brain. Workshop paper: Real Neurons and Hidden Units. Neural Information Processing Systems, 2019.

16. P Kürschner[†], S Dolgov, KD Harris, P Benner. [Greedy low-rank algorithm for smooth connectome regression](#). Journal of Mathematical Neuroscience 9: 9, 2019.
15. J Knox, KD Harris, N Graddis, JD Whitesell, H Zeng, JA Harris, E Shea-Brown, S Mihalas. [High resolution data-driven model of the mouse connectome](#). Network Neuroscience 3(1), 217-236, 2019.
14. KD Harris, T Dashevskiy, J Mendoza, AJ Garcia III, J-M Ramirez, E Shea-Brown. [Different roles for inhibition in the rhythm-generating respiratory network](#). Journal of Neurophysiology 118(4), 2070–2088, 2017.
13. A Litwin-Kumar, KD Harris, R Axel, H Sompolinsky, LF Abbott. [Optimal degrees of synaptic connectivity](#). Neuron 93, 1153–1164, 2017.
12. KD Harris, S Mihalas, E Shea-Brown. [High resolution neural connectivity from incomplete tracing data using nonnegative spline regression](#). Neural Information Processing Systems, 2016.
11. PS Dodds, EM Clark, S Desu, MR Frank, AJ Reagan, JR Williams, L Mitchell, KD Harris, IM Kloumann, JP Bagrow, K Megerdooimian, MT McMahon, BF Tivnan, CM Danforth. [Human language reveals a universal positivity bias](#). Proceedings of the National Academy of Sciences 112(8), 2389–2394. 2015.
10. KD Harris, PS Dodds, CM Danforth. [Dynamical influence processes on networks: General theory and applications to social contagion](#). Physical Review E 88, 022816, 2013.
9. L Mitchell, MR Frank, KD Harris, PS Dodds, CM Danforth. [The Geography of Happiness: Connecting Twitter sentiment and expression, demographics, and objective characteristics of place](#). PLoS ONE 8(5): e64417, 2013.
8. PS Dodds, KD Harris, CM Danforth. [Limited Imitation Contagion on Random Networks: Chaos, Universality, and Unpredictability](#). Physical Review Letters 110, 158701, 2013.
7. CA Bliss, IM Kloumann, KD Harris, CM Danforth, PS Dodds. [Twitter reciprocal reply networks exhibit assortativity with respect to happiness](#). Journal of Computational Science 3(5), 388–397, 2012.
6. KD Harris, E-H Ridouane, DL Hitt, CM Danforth. [Predicting flow reversals in chaotic natural convection using data assimilation](#). Tellus A 64, 17598, 2012.
5. N Allgaier, KD Harris, CM Danforth. [Empirical Correction of a Toy Climate Model](#). Physical Review E 85, 026201, 2012.
4. IM Kloumann, CM Danforth, KD Harris, CA Bliss, PS Dodds. [Positivity of the English language](#). PLoS ONE 7(1): e29484, 2012.
3. PS Dodds, KD Harris, IM Kloumann, CA Bliss, CM Danforth. [Temporal patterns of happiness and information in a global social network: Hedonometrics and Twitter](#). PLoS ONE 6(12): e26752, 2011.
2. JL Payne, KD Harris, PS Dodds. [Exact solutions for social and biological contagion models on mixed directed and undirected, degree-correlated random networks](#). Physical Review E 84, 016110, 2011.
1. PS Dodds, KD Harris, JL Payne. [Direct, physically motivated derivation of the contagion condition for spreading processes on generalized random networks](#). Physical Review E 83, 056122, 2011.

GRANTS AND HONORS

Washington Research Foundation Postdoctoral Fellowship, 4 yr. grant (used 3), 2018–2020

Graduate

- Boeing Research Award in Applied Mathematics, University of Washington, 2016
- Big Data for Genomics and Neuroscience Training Grant, University of Washington, 2015–2017
- Joseph Hammack Research Award in Applied Mathematics, University of Washington, 2014
- Boeing Fellowship, University of Washington, 3 yrs. funding, 2012–2015
- John F. Kenney Award for Excellence in Mathematics, University of Vermont, 2012
- NASA Graduate Research Assistantship, University of Vermont, 2009, 2011

Fulbright Scholar, Chile, 2010

- Worked with transportation engineers to optimize the nighttime bus timetables of Santiago

Undergraduate

- Vermont Scholar, funded 100% financial need, 2005–2009
- URECA! NASA-funded undergraduate research grant, 2008–2009
- Honors College Scholar, *Magna cum laude*, Phi Beta Kappa
- Senior Mathematics Award for Research, 2009

EDUCATION

University of Washington, Seattle, Washington.

- Ph.D., Applied Mathematics, December 2017.
Thesis: [This Brain Is a Mess: Inference, Random Graphs, and Biophysics to Disentangle Neuronal Networks](#). Advisor: Eric Shea-Brown
- Committee: Ioana Dumitriu, Adrienne Fairhall, Stefan Mihalas, Jan-Marino Ramirez

University of Vermont, Burlington, Vermont.

- M.S., Mathematics, October 2012.
Thesis: [On-off Threshold Models of Social Contagion](#).
Advisor: Peter Sheridan Dodds.
- B.A., Mathematics and B.A., Physics, May 2009.
Honors College thesis: [Predicting Climate Regime Change in Chaotic Convection](#).
Advisor: Christopher M. Danforth.

Other education

- Methods in Computational Neuroscience. Marine Biological Lab, Woods Hole, MA. 2015.
- Mining and modeling of neuroscience data. Redwood Center for Theoretical Neuroscience. University of California Berkeley, 2013.
- Complex Systems Summer School. Santa Fe Institute, 2011.
- ISEP Exchange Program. Pontificia Universidad Católica de Valparaíso, Chile. 2007.

TEACHING & MENTORING

Instructor

- Graduate Analysis of Algorithms (CSCI 511)
- Computational Neuroscience (CSCI 497V/597V)
- Machine Learning I (DATA 371)
- Analysis of Algorithms II (CSCI 405)
- Analysis of Algorithms I (CSCI 305)
- [CNeuro2022](#). Theoretical and Computational Neuroscience Summer School. IOB (Basel, Switzerland) and Tsinghua University (Beijing, China) S22
- Machine Learning (CSCI 471/571) F20
- Data Science for Biologists (UW BIO 419/519) W19
- [Introduction to Computational Models in Biology](#) (UW AMATH 422/522) F17

Research mentoring (WWU students unless otherwise noted)

- Juan Carlos Muratalla Sanchez, CS master's. 2025–
- Bob Gebert, CS master's. 2026–
- Mitchell Dohrman, CS master's. 2026–
- Emma Berreth, Math master's. 2026–
- Jeffrey Guo, CS undergrad. 2025–
- Jade Sweet, CS undergrad. 2024–2026.
- Ruchi Gupta, CS undergrad. 2025–2026.
- Pranav Cheraku, CS undergrad. 2025–2026.
- Victoria Hahn, Math undergrad. 2024–2026.
- Cameron Henderson, CS master's. 2024–2026.
- Joe Ewert, CS master's. 2024–2026.
- Robin Preble, CS master's. 2024–2026.
- Vivian White (→UBC CS PhD), CS undergrad & IN-BIC fellow in Montréal. 2022–2025.
- Rory Bates, CS undergrad. 2024–2025.
- Jed Pagcaliwagan, CS undergrad. 2024–2025.
- Josh Sonnen, CS undergrad. 2024–2025.
- Mayla Ward, CS & Math undergrad. 2024–2025.
- John-Paul Powers, CS grad. 2023–2024.
- Angus Read, CS undergrad & grad. 2021–2023.
- Cameron Kaminski (→Purdue), CS undergrad. 2022–2023.
- Suyhun “Michael” Ban, CS undergrad. 2021–2023.
- Caitlin Bannister (→Brown/NIH PhD program), neuroscience undergrad. 2021–2023.
- Jessica Stillwell (→PNNL), CS undergrad. 2020–2022.
- Grant Chou (→[Tuthill lab](#), Princeton PhD), CS undergrad. 2020–2022.
- Biraj Pandey, UW Applied Math PhD. 2019–2022.
- Sean McCulloch (→Allen Institute for Brain Science), CS master's. 2020–2021.
- Seth Hirsh (→Facebook), UW Physics PhD. 2018–2020.

- Satpreet Singh, UW Electrical & Computer Engineering PhD. 2018–2019.
- Yuchen Wang (→Adobe), UW CS & Engineering undergrad. 2018–2019.
- Nathan Lee, UW Applied Math PhD. 2018–2019.
- Joseph Knox (→Facebook), Allen Institute for Brain Science. 2017–2018.
- Nile Graddis, Allen Institute for Brain Science. 2015–2018.
- Joshua Mendoza (→PNNL). UW Applied Math Honors thesis: *The effects of network structure in creating a two-phase breathing pattern in the Bötzing and pre-Bötzing complexes*, 2014–2015.
- UWIN Postbac Fellowship, 2016.

Teaching assistant

- [Summer Workshop on the Dynamic Brain](#) (UW/Allen Institute for Brain Science) S16
- Women in Science and Engineering undergraduate program, UW S14
- STEM Bridge undergraduate program, UW S13
- Introduction to Nonlinear Dynamics and Chaos (UW AMATH 402/502) W13, W14
- Fundamentals of Calculus I (UVM MATH 19) F'08
- Basic Combinatorial Theory (UVM MATH 173) S'07

Instructor, Champlain Valley Union High School ACCESS community classes

- Intermediate Spanish: Advanced S12
- Spanish for Beginners F11

Guest lecturer

- Computational Modeling of Biological Systems (UW AMATH 422/522) F16
- Introduction to Nonlinear Dynamics and Chaos (UW AMATH 402/502) W15
- Linear Algebra (UVM MATH 124) F09
- Introduction to Numerical Analysis (UVM MATH 237) F09
- Chaos and Fractals (UVM MATH 266) S09

SERVICE

WWU Committees: Faculty senate (2023–2026), Undergraduate Curriculum (2023–, chair 2025–) Graduate (2021–2023), Executive (2022–2023). Library and Outreach (2025–2026).

Conference organizer: [DeepMath](#) (2022–); COSYNE 2024

Referee: CCN (2026), ICLR Sci4DL (2026), AISTATS (2025), ICML (2024, 2025), COSYNE (2021, 2022), NeurIPS (2021, 2022, 2024, 2025, 2026), SIAM Workshop on Network Science (2018), Network Neuroscience, NeuroImage, Physical Review Letters, PLOS ONE, PLOS Comp Bio, Physical Review E, New Media and Society, Neural Networks, Electronic Journal of Probability, Applied Mathematics and Computation

Volunteer, SIAM Math Fair at Lockwood Elementary School, 2014

UW computational neuroscience journal club co-organizer, 2014–2016

TALKS

Learning stochastic rainbow networks. DeepMath 2024, University of Pennsylvania.

What can we learn from random feature networks? University of Vermont, Complex Networks Seminar, August 2024

Using mathematics to untangle networks.

- WWU, Mathematics Colloquium, March 2024
- University of Oregon (Eugene), Department of Mathematics, January 2024

Functional geometry of biological structure. NeuroAI Montreal, October 2023

Geometry of neural networks with biological structure: Randomness vs. structure and learning.

- Montréal institute for learning algorithms (Mila) NeuroAI seminar. Montréal, QC. 2023.
- [CAIDA AI seminar](#), University of British Columbia. Vancouver, BC. 2022.
- Columbia University Center for Theoretical Neuroscience Seminar, February 2022

Network geometry for sensing and learning. NeuroAI Seattle, September 2022.

Computational Neuroscience. WWU Neuroscience Research Driven Students club, November 2021.

Randomness versus structure in networks.

- Center for Integrative Brain Research, Seattle Children's Research Institute, Seattle, WA. 2021.

- Computational Neuroscience Seminar, University of Washington, Seattle, WA. 2021.
- Mathematics Colloquium, Western Washington University, Bellingham, WA. 2021.

Leveraging the lacuna: spectral gaps and tensor recovery. Minisymposium: Latest advances in spectral linear algebra in network science. SIAM Linear Algebra. 2021.

Kernel theories of networks and their use in neuroscience. Computational Neuroscience Seminar, University of Washington, Seattle, WA. 2020.

Tensor complexity and completion using hypergraph expanders. Theory seminar. Computer Science & Engineering, University of Washington, Seattle, WA. 2020.

Brain network reconstruction as an inverse problem. Workshop: Inverse Problems in Imaging. SIAM PNW 2019, Seattle, WA. 2019.

Spectral gap in random bipartite biregular graphs and applications. Theory seminar. Computer Science & Engineering, University of Washington, Seattle, WA. 2019.

The Mouse Mesoscale Connectome: Data-driven Models and Organization of Cortical Networks. Showcase Symposium (team talk). Allen Institute for Brain Science, Seattle, WA. 2018.

Connections between dimensionality and network sparsity. Workshop: How does learning reshape dimensionality of collective network activity? Computational Neuroscience (CNS) 2018, Seattle, WA. 2018.

Advances and challenges in connectome inference and analysis.

- Institute for Disease Modeling, Bellevue, WA. 2017.
- Pacific Northwest National Laboratory, Richland, WA. 2017.

Construction of a voxel-based mesoscopic mouse connectome. Synergy Lecture. Allen Institute for Brain Science, Seattle, WA. 2016.

Role and limits of inhibition in an excitatory burst generator. Connecting Network Architecture and Network Computation. BIRS, Banff, AB. 2015.

Combined Effects of Connectivity and Inhibition in a Model of Breathing Rhythmogenesis. SIAM Dynamical Systems Conference. Snowbird Resort, Little Cottonwood Canyon, UT. 2015.

Limited Imitation Social Contagion as a Model of Fashions.

- Dynamics Days 2013. Marriott City Center, Denver, CO. 2013.
- SIAM Dynamical Systems Conference. Snowbird Resort, Little Cottonwood Canyon, UT. 2013.

Timetable optimization for Transantiago night service—Fulbright studies in Chile. Transportation Research Center Brown Bag Series. University of Vermont, Burlington, VT. 2011.

Geographical variation of happiness as expressed by users of Twitter. Student Research Conference. University of Vermont, Burlington, VT. 2011.

Programación de servicios nocturnos—Proponiendo un horario optimizado para Transantiago (Programming nightly bus service—Proposing an optimized schedule for Transantiago). Seminar “Herramientas avanzadas para la ciudad del futuro.” Universidad de Chile, Santiago, Chile. 2010.

Forecasting Flow Reversals in a Chaotic Toy Climate.

- SIAM Dynamical Systems Conference. Snowbird Resort, Little Cottonwood Canyon, UT. 2011.
- Applied Mathematics Seminar. University of Vermont, Burlington, VT. 2009.
- Graduate Student Seminar. University of Vermont, Burlington, VT. 2009.
- Student Research Conference. University of Vermont, Burlington, VT. 2009.
- MAA Northeastern Sectional Meeting. St. Michael’s College, Burlington, VT. 2008.
- URECA! Awards Ceremony. University of Vermont, Burlington, VT. 2008.

PROFESSIONAL DEVELOPMENT

BETTER in STEM teacher training, 2024–2026 cohort

TECHNICAL SKILLS

Programming languages: Python (pytorch, tensorflow, scipy, numpy, networkx, scikit-learn, pandas), MATLAB, C/C++

Data: Experience with PostgreSQL + PostGIS, Hadoop + Pig, R

Computer systems: cluster environments, Linux, L^AT_EX, git

Spoken language: Spanish (fluent), English (native), French (basic)

REFERENCES

Bingni Wen Brunton

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